

DR KIRAN KUMAR KONDRU

OBJECTIVE

Recent Computer Science Ph.D. graduate specializing in distributed systems, seeking a teaching position to blend cutting-edge research with innovative pedagogy. Committed to foster a dynamic learning environment that emphasizes practical application, critical thinking, and collaborative problem-solving. Eager to inspire the next generation of computer scientists by bridging theoretical concepts with real-world challenges, particularly in areas such as blockchain technology and distributed computing. Aiming to contribute to the academic community through a balance of impactful research and student-centered teaching methodologies.

EDUCATION

Ph.D. in Computer Science

2017 - 2024

Central University of Tamil Nadu

Thiruvavur, TN

- Advisor: Dr R. Saranya
- Thesis Title: Re-Engineering the Raft Distributed Consensus Algorithm with enhanced Fault-tolerance and improved Scalability

Master of Computer Applications

2003 - 2006

Hindu College P.G. Courses

Guntur, AP

- Affiliated to Acharya Nagarjuna University, Guntur

Bachelor of Computer Applications

2000 - 2003

Jagaramudi Kuppuswamy Choudary College

Guntur, AP

- Affiliated to Acharya Nagarjuna University, Guntur

RESEARCH EXPERIENCE

Scalable Raft Implementation with Proxy Leaders | Advisor: Dr R Saranya

- Developed and implemented RaftOptima, an enhanced Raft algorithm addressing scalability issues in large clusters. Introduced proxy leaders to delegate broadcasting and acknowledgment tasks, reducing the latency by 60% in clusters of up to 25 nodes. Constructed a key-value data store and network simulator to validate the system, demonstrating significant throughput improvements while preserving core safety properties. Delivered comprehensive reports and presentations summarizing findings.

Development of Enhanced Raft Leadership Algorithm | Advisor: Dr R Saranya

- Developed RaftOptima, a Java-based enhanced Raft consensus algorithm addressing availability issues by resolving symmetric and transitive reachability problems that triggered frequent leader elections. Built a key-value store for replication testing and a custom Discrete Event Simulator to benchmark performance under network partitions, achieving reduced downtime and improved cluster availability. Conducted formal verification of leader election safety using TLA+ to ensure algorithmic correctness under failure scenarios

PUBLICATIONS

1. **Kondru, Kiran Kumar** and Saranya Rajiakodi. "RaftOptima: An Optimised Raft with enhanced Fault Tolerance, and increased Scalability with low latency." *IEEE Access* (2024). DOI: 10.1109/ACCESS.2024.3435978
2. **Kondru, Kiran Kumar** and R. Saranya. "Exploring the Limits of Raft's Fault Tolerance: Insights from Simulated Network Partitions," *International Journal of Intelligent Systems and Applications in Engineering*, vol. 12, no. 22S, pp. 838–850, 2024..
3. **Kondru, Kiran Kumar** and R. Saranya. "Directed acyclic graph-based distributed ledgers—an evolutionary perspective." *Int J Eng Adv Technol (IJEAT)* 9.1 (2019), DOI: 10.35940/ijeat.A1970.109119.

BOOK CHAPTERS	1. Kondru, Kiran Kumar , R. Saranya and Annamma Chacko. "A Review of distributed supercomputing platforms using Blockchain." <i>Advances in Distributed Computing and Machine Learning: Proceedings of ICADCML 2020</i> (2021): 123-133, DOI: 10.1007/978-981-15-4218-3_13
CONFERENCES	1. Kondru, Kiran Kumar and Saranya Rajiakodi. "Improving scalability of permissioned blockchains by making Raft Orderer to understand the underlying network topology." <i>Proceedings of the 1st International Conference on Artificial Intelligence, Communication, IoT, Data Engineering and Security, IACIDS 2023</i>, 23-25 November 2023, Lavasa, Pune, India. 2024, DOI: 10.4108/eai.23-11-2023.2343202 2. Evanjalín, A. Beulah, R. Saranya, and Kiran Kumar, Kondru. "Illustration of a Parallel and Discrete Event Simulator using Scala." 2021 First <i>International Conference on Advances in Computing and Future Communication Technologies (ICACFCT)</i>. IEEE, 2021, DOI: 10.1109/ICACFCT53978.2021.9837380. 3. Kiran Kumar, Kondru and Dr R Saranya. K. K. Kumar and D. R. Saranya, "The Evolution of Blockchain – From Chain of Blocks to Direct Acyclic Graphs," in <i>National Conference on Blockchain and Smart Contract Technologies (BSCIT 2019)</i>, Dr. K. Singh, Ed., National Conference on Blockchain and Smart Contract Technologies, Nov 7-10, 2019, pp. 18–23.
INDUSTRY EXPERIENCE	Progress Software Development LTd Software Engineer II Apr 2009 - Dec 2009 • As a Software Engineer II, I contributed to the development of Progress® OpenEdge® Architect, an Eclipse-based integrated development environment for services-oriented business applications. My responsibilities included designing and developing Eclipse plugins to support Advanced Business Language (ABL), enhancing existing IDE features, and building prototypes such as an SWT/Swing bridge for the Apama Event Modeller. I also implemented test cases for core modules and extended OpenEdge's Open Client, ensuring robust and extensible development tools for business logic and application data management. Commvault Systems(INDIA) Pvt Ltd Member, Technical Staff Mar 2007 - Apr 2009 • As a Member of Technical Staff in the development team, I focused on designing and implementing graphical user interfaces for new features in the Simpana Suite, a unified data protection solution. My role involved developing Java Swing-based GUIs, establishing communication protocols between client and server, and ensuring the quality of new features through unit testing and validation. I led the implementation of an XML-based protocol, built wizard-based GUI prototypes, and mentored new team members, delivering solutions for major clients including Fidelity, Royal Dutch Shell, and the US Air Force.
PROFESSIONAL DEVELOPMENT	2010 - 2015: Following my experience in the IT industry, I embarked on a period of intensive career exploration and personal growth. During this time, I rigorously prepared for Civil Services examinations, significantly enhancing my analytical skills, time management, and broad knowledge of Indian and global affairs. This journey culminated in my successful clearance of the UGC NET with JRF in 2015, demonstrating my strong academic aptitude and commitment to pursuing a career in academia. This transformative period honed my critical thinking, research methodologies, and adaptability, while deepening my understanding of complex subjects across various fields. This unique combination of industry knowledge and academic rigour has prepared me exceptionally well for a dynamic role in computer science education and research
AWARDS AND FELLOWSHIPS	• NET JRF, University Grants Commission (UGC), 2015 • NET for Asst. Professor, University Grants Commission (UGC), 2014 • APSET for Asst. Professor, Govt of AP, 2013

SKILLS	Research: Advanced Research Methodologies, Data Analysis and Interpretation, Academic writing and publication, Project Management, Critical thinking and problem-solving. Teaching: Curriculum development, Lecture preparation and delivery, Student mentoring and advising, classroom management Soft Skills: Effective communication(verbal and written), Leadership and team collaboration, Time management and organization, Adaptability and continuous learning, Cultural sensitivity and inclusivity Programming: C, Java, Python.
REFERENCES	Dr R. Saranya: Supervisor and Assistant Professor. <i>Department of Computer Science, Central University of Tamil Nadu, Thiruvavur, India</i> (saranya@acad.cutn.ac.in) Dr A. Martin Member, Research Advisory Committee and Assistant Professor. <i>Department of Computer Science, Central University of Tamil Nadu, Thiruvavur, India</i> (martin@cutn.ac.in) Dr. Chandra Mouli P.V.S.S.R Head of the Department, Professor and Dean. <i>Department of Computer Science, Central University of Tamil Nadu, Thiruvavur, India</i> (chandramouli@cutn.ac.in)